SDC HACKATHON 2026

Phase 1: Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

1. Team & Project Overview

Team Name	**JJ**
Selected Domain	**Orbyte Smart SME Challenge:Less Busywork, Better Business**
Selected SDG Track (1–17)	**SDG 8**
Project / Solution Name	**ESG-Aware Receivables Risk & Collection Prioritization for Malaysian SMEs**
Name (Leader)	**LYU QIUYAO**
Email & Phone (Leader)	**dsc2409023@xmu.edu.my & +8618323302858**
University Affiliation(s)	**xiamen university malaysia**
Number of Team Members	**2**

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Malaysian small and medium enterprises face an average payment delay of 64 days, 6 days longer than large corporates, with the gap widening to 26 days in logistics and hospitality sectors; micro enterprises can face overdue periods as long as 120 days. MSMEs make up more than 97% of all business establishments in Malaysia, employing over two-thirds of the national workforce, yet most manage receivables manually across Excel files, WhatsApp chats and scattered emails.

When multiple invoices approach or pass due dates, finance staff must manually cross-check overdue days, outstanding balances, customer promises and disputes across disconnected tools, leading to duplicated follow-ups, missed reminders and delayed cash-flow warnings. Adding ESG or supplier sustainability records to credit decisions further raises judgement costs, with no central, auditable trail of decisions. Cash-flow pressure from delayed receivables directly threatens business survival, stable employment, and supplier stability across local supply chains; weak receivables management also makes MSMEs unreliable trading partners in sustainable regional supply networks.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

We build a runnable web application for MSME finance administrators, addressing the receivables pain points above by combining deterministic rule engines with targeted AI assistance. Users upload customer correspondence through a single intake channel, and AI automatically matches text to the correct company, then uses NLP to extract overdue days, outstanding amounts, follow-up items, customer promises and dispute points, with all extracted values cited directly to source text. Ambiguous content never gets auto-filed; administrators manually confirm attribution before the system generates company profiles with predefined risk tags.

The workflow is driven by transparent rules: customers with existing ESG ratings enter a three-factor weighted queue (overdue days, outstanding amount, ESG score), while customers without ESG ratings enter a two-factor queue for step-by-step human review. AI only provides cash-flow explanations and conditional action suggestions, never modifying core amounts, scores or queue order. After administrators select immediate collection or negotiated extension, AI pulls corresponding English and Bahasa Melayu paragraphs from a controlled sentence library, which staff can freely edit before explicit send confirmation.

We leverage skills from SDC Phase 1 workshops: Replit-hosted generative AI APIs for NLP extraction and profile generation, structured PostgreSQL database architecture with Zod validation, and Replit cloud deployment. All AI calls fall back to deterministic rules and templates on failure, with append-only audit logs tracking every action.

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

• Frontend: React + TypeScript, Tailwind CSS + shadcn/ui components, Vite for build, Recharts for dashboard charts • Backend: Node.js + Express 5, Zod for end-to-end request validation, structured runtime logging • Database: PostgreSQL with Drizzle ORM; current demo version uses an in-memory repository store, with hosted PostgreSQL planned for production deployment • AI: Replit-managed OpenAI GPT model (server-side) for customer matching, NLP correspondence extraction, company profile generation, and bilingual notice paragraph selection; all AI calls fall back to deterministic rules and templates on timeout or failure • Deployment: Full stack hosted and run on the Replit cloud platform, with esbuild for API bundling and Vite for frontend build We used SDC’s sponsored tools: the Replit-hosted managed OpenAI Generative AI API, and Replit cloud hosting for end-to-end development and deployment. No private API keys are hardcoded, and all AI calls are run server-side with platform-managed credentials.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We decomposed the administrator’s real receivables workflow into six focused screens: dashboard, correspondence intake, risk detection, manual review, execution results, and audit logs, so finance staff never need to toggle across disconnected files. AI acts only as an assistant: core actions including amount calculation, scoring, queue ordering, approval and sending are controlled by deterministic rules and human judgement; if AI calls fail, rule outputs remain valid. Ambiguous correspondence is never auto-filed, and all pre-send content is reviewed by staff. Without configured SMTP, the system runs in simulation mode, letting new users practice the full workflow with zero risk of accidental real sends. Weights, rule snapshots and bilingual drafts are shown directly in the UI, so every action is explainable rather than blindly trusted. For Phase 2, we will migrate the in-memory demo store to hosted PostgreSQL, support customizable scoring weights, and address the issue that merchants can only upload text files for company profile extraction, with video uploads unsupported.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC’s focus on meaningful global sustainability, establish your project’s impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess [Innovation and Creativity as well as Feasibility of build]. Vague or unsupported projections may weaken your case for advancing to Phase 2.

Metric Category	**Your Response**
Target SDG Sub-Target(s)	Primary: SDG Target 8.3 − Promote development-oriented policies that support productive activities, decent job creation,

Metric Category	Your Response
Identify the exact UN SDG sub-target your project addresses *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	entrepreneurship, creativity and innovation, and encourage the formalization and growth of micro-, small- and medium-sized enterprises. Secondary: SDG Target 8.10 − Strengthen the capacity of domestic financial institutions to encourage and expand access to banking, insurance and financial services for all.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Malaysian SMEs face an average payment delay of 64 days, 6 days longer than large corporates, with the gap widening to 26 days in logistics and hospitality sectors; micro enterprises can face overdue periods up to 120 days. Local industry case data shows MSME finance teams spend an average of 15 hours per month manually tracking receivables across WhatsApp, email and Excel spreadsheets. World Bank 2021 business survey data finds 38% of Malaysian MSMEs hold less than 1 month of operating cash, with delayed receivables as the leading cash-flow pressure.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	1) Monthly manual receivables follow-up hours per finance team; 2) Average customer payment delay days; 3) Customer review completion rate; 4) English/Malay bilingual notice coverage; 5) Pre-send human confirmation rate; 6) Audit event capture rate.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	91% reduction in manual receivables follow-up workload compared to traditional Excel/WhatsApp manual processes; 20% reduction in average customer payment delay (from the current 64-day SME baseline to below 51 days); 100% customer review, bilingual coverage and pre-send confirmation rates; 100% audit event capture rate;

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React / React DOM	Code (frontend framework)	MIT
Vite	Code (build tool)	MIT
Tailwind CSS	Code (styling)	MIT
Radix UI / shadcn/ui components	Code (UI components)	MIT
TanStack Query	Code (data fetching)	MIT

Recharts	Code (charts)	MIT
Express	Code (web framework)	MIT
Zod	Code (validation)	MIT
Drizzle ORM	Code (ORM)	Apache-2.0
PostgreSQL	Code / database	PostgreSQL License
Nodemailer	Code (email)	MIT
OpenAI Node SDK	Code (AI client)	Apache-2.0
pino	Code (logging)	MIT
esbuild	Code (build)	MIT
OpenAI GPT model (via Replit AI Integrations)	Model	OpenAI API Terms
Synthetic demo dataset (10 companies / 26 invoices / 8 ESG records)	Dataset	Team-created, no third-party data
Replit platform (development,hosting, AI integration)	Cloud platform	Replit Terms of Service

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

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7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature] LYU QIUYAO

[Date] 2026.10.24